

but its ordered correctly. you can see it in my notebook. what should i do

I think we've narrowed it down to the real issue.

The key contradiction is this:

- `fare_per_mile` still has **76** infinite values.
- `trip_distance <= 0` also returns **76**.

If your filtering had actually been applied to the `taxi` DataFrame that you later use, then **both of these numbers should be 0**.

That means one of two things is happening:

1. The filtering cell never actually modified `taxi` (for example, it wasn't assigned back to `taxi`).
2. You're checking or using a different `taxi` DataFrame than the one you filtered.

Let's verify that with one simple check.

Right after your filtering line, run these two commands:

<> Python

```
print(taxi.shape)
print((taxi["trip_distance"] <= 0).sum())
```



Run

The most likely explanation now

Since your code is correct but your outputs still say:

```
trip_distance <= 0 : 76
fare_per_mile inf : 76
```



I suspect the notebook is not running the cells you think it is.

Do this exactly:

1. Kernel → Restart Kernel and Clear Outputs
2. Run **only** the preprocessing cell.
3. Don't run anything else.
4. Tell me the output of these three lines:

<> Python

```
print(taxi.shape)
print((taxi["trip_distance"] <= 0).sum())
print(np.isinf(taxi["fare_per_mile"]).sum())
```



Run

```

-----
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_27575/718584144.py in <cell line: 0>()
      4
      5 linear_model = LinearRegression()
----> 6 linear_model.fit(X_train, y_train)
      7
      8 train_pred = linear_model.predict(X_train)

6 frames
Show more ▾

```

At this point, we shouldn't guess. We need to identify **exactly** what's in `X_train`.

Run these four cells **before** `linear_model.fit(...)`:

1. Check for infinity

```

Python
import numpy as np

print("Infinity values:", np.isinf(X_train).sum().sum())

```

```

Rows BEFORE cleaning: (41202, 15)
Rows AFTER cleaning: (41126, 15)
Zero distances: 0
Min distance: 0.01

```

Excellent! That means the cleaning is now working correctly:

- ✓ Rows before cleaning: **41,202**
- ✓ Rows after cleaning: **41,126**
- ✓ Zero-distance trips: **0**
- ✓ Minimum trip distance: **0.01**

So the preprocessing is **fixed**. 🎉

The reason you're **still** getting the error is that the later variables (`taxi_encoded`, `X_train`, etc.) were created **before** you fixed the preprocessing. They still contain the old data with `inf` values.

```
[45010.96663458451, 41483.1080910755, 39253.69647655582,
37126.341195815694, 34553.30296651279, 32667.47346019493,
30564.067929276433, 28749.843659710354,
26914.819859896274, 26914.819859896274] is inertia
```

I can see the problem now.

Your `inertia` list is:

Python



Run

```
[
  45010.96,
  41483.10,
  39253.69,
  37126.34,
  34553.30,
  32667.47,
  30564.07,
  28749.84,
  26914.82,
  26914.82
]
```

It has **10 values**, but your x-axis is:

```
for k in range(2,11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init = 10)
    kmeans.fit(X_scaled)
    inertia.append(kmeans.inertia_)
```

Your loop is **correct**.

The problem is that you **didn't clear** `inertia` **before running the cell again**.

Notice your list:

Python



Run

```
[
  45010.96,
  41483.10,
  39253.69,
  37126.34,
  34553.30,
  32667.47,
  30564.07,
  28749.84,
  26914.82,
  26914.82 ← duplicate last value
]
```

The last value appears twice.

